Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **distributed data platforms**, **Java/SQL applications**, cloud ETL pipelines, and **fault-tolerant analytics workflows** over large datasets.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Data Retrieval, Object-Oriented Programming

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant distributed services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load.
- Achieved 99%+ reliability across 5K+ monthly jobs through **incremental data pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production data platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive analytical workflows during concurrent model execution and sustained production traffic reliably.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **quantitative modeling pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and reproducible evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal model** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Increased pipeline efficiency by 68% through **AWS Glue and S3 ETL**, transforming JSON, CSV, and Parquet inputs into normalized downstream datasets while reducing storage costs by 15% reliably.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Cyber Security and Data Mining: Detecting Malicious URL **May 2023 – August 2023**
Data Analyst - Research Fellowship | *India*
- Achieved 96%+ classification accuracy across 650K+ URLs by comparing **Random Forest, CNN, RNN, LSTM, Bi-LSTM, and GRU models** within a reproducible analytical pipeline for malicious-pattern detection reliably.
- Engineered 21 behavioral indicators covering URL length, IP presence, and suspicious keywords, translating noisy inputs into **classification features** that exposed phishing and malware patterns reliably at scale.

Skills

Languages: Java, SQL, Python, C++, TypeScript, JavaScript, Bash
Big Data: Spark, Databricks, Hadoop, AWS Glue, AWS S3, ETL, Parquet, Data Pipelines, Distributed Computing
Distributed Systems: Fault Tolerance, Failure Isolation, Consistency, Availability, Durability, Concurrency, Multithreading, Caching, Performance Optimization
Data & Cloud: AWS, Kubernetes, Docker, PostgreSQL, MySQL, MongoDB, AWS RDS, Schema Design, Query Optimization
Engineering & Quality: REST APIs, Object-Oriented Programming, Data Structures, Algorithms, Unit Testing, Integration Testing, PyTest, Mockito, Git, GitHub, Jenkins, CI/CD, Linux, Debugging, Observability

Projects

Cloud Lakehouse Processing Pipeline - Design Concept
- Designed a **cloud lakehouse pipeline** concept ingesting batch and streaming records into S3, transforming Parquet datasets with Spark, validating schemas, isolating tenants, tracking lineage, and supporting reproducible backfills.

Distributed Analytics Service - Design Concept
- Designed a **distributed analytics service** concept with Java APIs, partitioned workloads, idempotent processing, Kubernetes deployment, retry policies, consistency checks, cached results, operational metrics, and fault-tolerant recovery workflows.

Chat2SQL | Live
- Unified **10+ database systems** in a Java and TypeScript workspace extending Chat2DB, integrating schema exploration, SQL generation, structure synchronization, and Docker-based local deployment across Windows, macOS, and Linux environments.